Faiz Hussain

AI Engineer | MLOps Engineer

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SUMMARY

AI and MLOps Engineer with 2+ years of production DevOps experience on AWS at NTT DATA and an MSc in Data Science from the University of Naples Federico II. I build and run ML and LLM systems end to end: experiment tracking in MLflow, serving with FastAPI and Docker, and deployment on Kubernetes with Terraform and CI/CD. Research in LLM security across four open-source models; portfolio work on multi-tenant RAG and release gating for model quality. EU work authorisation.

EXPERIENCE

MSc Thesis Researcher – LLM Security | University of Naples Federico II, Naples — Jan 2026 – Sep 2026

- Reproduced the Cloak, Honey, Trap proactive defences against LLM agents (USENIX Security 2025) and extended them to **4 locally hosted 4-bit LLMs** (Llama-3.2 1B and 3B, Mistral-7B, Qwen2.5-1.5B); supervised by Prof. Giancarlo Sperli.
- Built a reproducible evaluation pipeline in Python and Pandas on **285 MMLU questions across 57 subjects**, measuring defence success rate per model, per subject and per injection count (1 to 15).
- Showed honeytoken detection at **100%** on every model, while Cloak and Trap weakened as models grew (**23.4%** Trap DSR on Llama-3.2-1B vs **3.9%** on Mistral-7B); stacking both defences was super-additive.

DevOps Engineer (Senior Associate, IT) | NTT DATA, Bangalore — Sep 2021 – Aug 2023

- Built and managed production AWS infrastructure (EC2, ECS, VPC, S3) with load balancing and auto-scaling.
- Automated infrastructure provisioning with Terraform, increasing deployment speed by **45%**.
- Developed CI/CD pipelines in Jenkins and AWS CodePipeline, cutting release time from **hours to minutes**.
- Deployed and maintained containerised microservices on Kubernetes (EKS) using Docker and Helm.
- Configured CloudWatch monitoring, logging and alerting; reduced AWS infrastructure costs by **20%** through right-sizing.

EDUCATION

MSc in Data Science | University of Naples Federico II, Italy — 2023 – 2026

BTech in Engineering | Dayananda Sagar University, India — 2017 – 2021

SKILLS

- **Cloud:** AWS (EC2, EKS, ECS, S3, Lambda, VPC, ECR, SageMaker, IAM)
- **MLOps:** MLflow, Kubeflow, SageMaker Pipelines, DVC, FastAPI
- **Infrastructure as Code:** Terraform, Helm, Ansible
- **Containers and Orchestration:** Docker, Kubernetes (EKS)
- **CI/CD:** GitHub Actions, GitLab CI, Jenkins, AWS CodePipeline
- **Monitoring:** Prometheus, Grafana, CloudWatch
- **Programming:** Python, SQL
- **Machine Learning and Data:** scikit-learn, PyTorch, Pandas, Apache Spark, PostgreSQL

PROJECTS

TrustGate – LLM Evaluation and Release Gating | GitHub — Aug 2026

- Built an evaluation pipeline that scores LLM, RAG and agent systems, estimates quality with confidence intervals, and returns **ship / investigate / block** through a statistical non-inferiority test.
- Runs as a **GitHub Action** that fails a pull request on a quality regression; prediction-powered inference matched classical precision with **~50% fewer human labels** in a 2,000-item simulation. **51 tests**.

Technologies: ***Python, GitHub Actions***

Multi-Tenant RAG Customer-Support Agent | Live · GitHub — Jul 2026

- Built a SaaS support agent that answers each tenant only from its own documents, with isolation enforced in SQL on every vector query (OWASP LLM08).
- Added confidence gating that escalates uncertain answers to a human; a concurrent load test with **3 tenants and 90 requests** showed **0 cross-tenant leaks** at **52.6 requests/s**.

Technologies: ***Python, FastAPI, PostgreSQL, Docker, GitHub Actions***

Kubernetes Log Anomaly Detection – MLOps Pipeline | GitHub — Jan 2026

- Built a train, score and monitor pipeline that ranks **11,000** Kubernetes log lines so on-call engineers review the top 50; every run, parameter and artifact is tracked in MLflow.
- Reached **ROC-AUC 0.88** and **precision@50 of 1.00** on synthetic data (in-sample); CI runs unit tests, an end-to-end smoke test and the full pipeline on every push.

Technologies: ***Python, scikit-learn, MLflow, GitHub Actions***

CERTIFICATIONS AND LANGUAGES

- **AWS Certified:** DevOps Engineer – Professional | Solutions Architect – Associate | Cloud Practitioner
- **Languages:** English (C1), Italian (A2)

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